

The effect of ocean alkalinity enhancement on zooplankton standing stock and community composition in the Eastern Mediterranean Sea: a mesocosm study

Ariadna C. Nocera^{a,*}, Claudia Traboni^b, Justine Courboulès^c, Filomena Romano^d, Christos Chantzaras^e, Iordanis Magiopoulos^{d,e}, Selene Varliero^f, Arianna Azzellino^g, Pietro Bazzicalupo^h, Piero Macchi^f, Guido Raos^f, Daniela Basso^h, Paraskevi Pitta^{d,**}

^a Centro para el Estudio de Sistemas Marinos (CCT CENPAT-CONICET), Puerto Madryn, Argentina

^b Stazione Zoologica Anton Dohrn, Integrated Marine Ecology, Naples, Italy

^c Norwegian Institute of Science and Technology, Biology, Trondheim, Norway

^d Hellenic Centre for Marine Research, Institute of Oceanography, Heraklion, Greece

^e University of Crete, Department of Biology, Heraklion, Greece

^f Politecnico di Milano, Department of Chemistry, Materials and Chemical Engineering "G. Natta", Milan, Italy

^g Politecnico di Milano, Department of Civil and Environmental Engineering, Milan, Italy

^h University of Milano-Bicocca, Department of Earth and Environmental Sciences, Milan, Italy

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ABSTRACT

Ocean acidification (OA) due to anthropogenic CO₂ emissions has significantly altered ocean chemistry since the industrial era. Ocean alkalinity enhancement (OAE) is an innovative strategy to mitigate excess CO₂, with ocean liming (OL) serving as a potential carbon dioxide removal (CDR) method, through the spreading of Ca(OH)₂ (slaked lime) at the ocean surface. This study examined the ecological effects of OL on a natural zooplankton community from the ultraoligotrophic Eastern Mediterranean Sea during a 14-day mesocosm experiment conducted in spring-summer. We investigated how varying concentrations of slaked lime (low: 0.00074 g L⁻¹, high: 0.0067 g L⁻¹) affected zooplankton community structure, abundance, composition and species diversity. The experiment revealed complex temporal dynamics in the zooplankton community across all treatments. Tintinnina and Copepoda dominated the community (80–100 %), with nauplii significantly outnumbering copepodites and adults by an order of magnitude. Redundancy Analysis and correlations showed that pH, temperature and nutrient availability were crucial factors shaping the zooplankton community structure. These environmental variables explained more of the community variation than the lime treatments themselves, highlighting the importance of considering multiple ecological factors when assessing OAE impacts. The overall community response to OL treatments was subtle, suggesting potential resilience of the zooplankton community to short-term alkalinity enhancement. This research provides insights into the ecological implications of OL as a potential OA mitigation strategy. It emphasizes the need for longer-term studies to fully understand the cascading effects on marine food webs and ecosystem functioning. Future research should focus on the interplay between OL, nutrient dynamics, and trophic interactions to better predict the ecological consequences of large-scale OAE implementation.

1. Introduction

In marine ecosystems, plankton is a key component contributing to the global biogeochemical carbon cycle, in a series of processes known

as the Biological Carbon Pumps (BCP; [Boyd, 2019](#)). The sequence starts at the ocean's surface where primary producers transform the dissolved inorganic carbon (DIC) into organic matter by means of photosynthesis ([Ducklow et al., 2001](#)). The energy can be either transferred to higher

* Corresponding author.

** Corresponding author.

E-mail addresses: arinocera@gmail.com, anocera@cenpat-conicet.gob.ar (A.C. Nocera).

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levels in the food web or exported to the sea bottom (Richardson and Jackson, 2007). Subsequently, carbon export to deeper layers is partially controlled by higher-order organisms through grazing and the release of fecal pellets (i.e., by zooplankton; Turner, 2015). It is also influenced by the particulate organic matter (POM) sedimentation of marine aggregates (living and dead phytoplankton cells, etc.), the addition of biomineral particles, bacteria, and viruses, as well as by the mixing of the dissolved organic fraction (Iversen, 2023). Finally, the sinking POM (formed from a combination of phytoplankton cells, zooplankton fecal pellets, and denser detritus) will sediment to the seafloor (Siegel et al., 2024), where it could be sequestered for extended periods preventing the re-injection of carbon into the system (Claustre et al., 2021).

The natural capacity of the global ocean to absorb CO₂ has diminished over the past centuries due to anthropogenic emissions, creating an imbalance between atmospheric and oceanic exchanges of this gas (Doney et al., 2009, 2024). However, nearly 30 % of the atmospheric CO₂ annually emitted by human activities is being absorbed by the ocean resulting in a reduction of seawater pH (Friedlingstein et al., 2022), causing the ocean acidification. Consequently, the BCP efficiency is being affected and modified with changes in ecosystems functioning and marine organisms such as lower energy transfer efficiency (Cripps et al., 2016; Algueró-Muñiz et al., 2017; Meyers et al., 2019), dissolution of carbonate structures (e.g., Hofmann et al., 2010), marine biodiversity loss (e.g., Guinotte and Fabry, 2008) and food web alterations (Alvarez-Fernandez et al., 2018; Hammill et al., 2018; Riebesell et al., 2018) among others.

Different carbon dioxide removal (CDR) strategies are presently being proposed, which encompass a range of methods aimed at reducing atmospheric CO₂ levels, by directly extracting CO₂ from the atmosphere or by enhancing land and ocean carbon sinks capacities to increase CO₂ sequestration from the atmosphere (Johnson et al., 2024). Coastal and ocean systems present promising options for storing excess carbon due to their vast areas and potential for long-term storage (e.g., Siegel et al., 2021; Mengis et al., 2023). Specifically, solutions or interventions that employ technical or artificial means to modify natural processes and influence carbon uptake in marine systems are referred to as marine CDR (mCDR) actions (Tamme and Beck, 2021; Oschlies et al., 2023). This is primarily achieved by enhancing the net air-sea CO₂ flux through various methods, such as boosting biological productivity or altering ocean chemistry (e.g., Gagern et al., 2022; Bach et al., 2024). Among these, ocean alkalinity enhancement (OAE) is being considered a promising pathway to mitigate ocean acidification by increasing seawater pH (Iglesias-Rodríguez et al., 2023).

Nevertheless, the efficacy and the extent of the potential effects of the OAE on marine ecosystems and their organisms are still understudied. Recent research efforts devoted to the study of OAE on natural plankton communities were conducted by Pedersen and Hansen (2003; planktonic community), Subhas et al. (2022; bacteria), Ferderer et al. (2022; bacteria and phytoplankton), and Guo et al. (2024; bacteria, phytoplankton and zooplankton). The specific effects on targeted species have also been studied (e.g., Camatti et al., 2024). The effects of OAE through olivine and/or lime addition on these model organisms and experimental systems revealed negligible response at the bacterial level, but a shift towards smaller phytoplankton cells (Subhas et al., 2022; Guo et al., 2024) with effects on higher trophic levels, such as mesozooplankton both in natural (Guo et al., 2024) and laboratory (Camatti et al., 2024) studies. In addition, enhanced ciliate biomass, but unchanged dinoflagellate abundance was recorded in more alkaline conditions (Pedersen and Hansen, 2003). However, these experiments were sometimes conducted using relatively small water volumes (e.g., microcosms), and the outcomes are rather discrepant, leaving a current gap of certainty about the effect of OAE on natural plankton communities. Although valuable, the present knowledge is not sufficiently robust to promote a large field implementation of OAE on the natural ecosystem at this stage. There is an immediate need to investigate the effects of OAE on a larger scale and longer time, particularly focusing on

key components of marine food webs, which all higher trophic levels depend on. Consequently, larger volumes, such as those provided by mesocosms, could enhance our understanding by offering a more accurate representation of realistic conditions to assess the potential effects of OAE on the entire plankton community, from viruses to mesozooplankton.

Here, we describe a study which was part of a mesocosm experiment, using a Cretan Sea coastal plankton community to investigate the OAE effects on the following: i) the dynamics of the zooplankton abundance and community composition at different alkalinity concentrations, ii) the possible zooplankton diversity decrement, iii) the ontogenetic shifts in specific zooplankton groups (copepods) as a function of the alkalization process. Our hypothesis supports the idea that, at ambient pH levels, changes in the micro-mesozooplankton community might result from the enclosure conditions, leading to resource exhaustion and promoting more opportunistic and flexible taxa.

2. Methods

2.1. Mesocosm set-up and experimental design

The mesocosm experiment was held in May–June 2023 at the CretaCosmos facility (<https://www.aquacosm.eu/mesocosms/creta-cosmos>) of the Hellenic Centre for Marine Research (HCMR), in Crete, Greece. A 150 m³ land-based concrete pool (4 m deep) was used for the incubation of mesocosm-bags; the pool had a continuous water flow with water pumped from a well into the pool, in order to keep the temperature stable, recreating similar natural conditions through the entire experiment period. The seawater for the mesocosm bags was collected on May 26th from a coastal location close to HCMR, approximately at 200 m offshore and at 1.5 m depth. The seawater pumped was placed in barrels made of high-density polyethylene (HDPE; ~1 m³), previously rinsed (5 % HCl and deionized water) and transported by trucks to the mesocosm facility. The seawater was then distributed homogeneously between 9 different mesocosms, where each mesocosm enclosed in total ~3 m³ of natural seawater. Mesocosm structures consisted of transparent polyethylene bags with a circular opening of 1.3 m diameter, and a height of 2.34 m, and with a plexiglass cover endowed with an opening used for sampling purposes. The bags were supported by thick metal rings through which they were anchored to the edge of the pool, located in the outside area of the HCMR; hence, the mesocosms maintained similar to the *in situ* light and temperature conditions. Finally, the mesocosm bags were left to settle on May 27th (here denoted as D₀), before the treatments commenced.

The experiment was carried out from May 28th (D₁) until June 12th (D₁₄; Fig. 1). Alkalization was reached through the addition of Ca(OH)₂, which was dosed as a 1.5 M concentration of slurry, prepared by mixing ~5 min the desired quantity of Ca(OH)₂ with water sampled from the selected mesocosm as the alkaline treatment. Six successive additions to corresponding mesocosm bags were performed every other day (Fig. 1), on days 1, 3, 5, 7, 9 and 11. Nine mesocosms were employed during the experiment as follows: three control mesocosms (C1, C2 and C3; without slaked lime; hereafter referred to as C-mesocosms), three low slurry concentration mesocosms (L1, L2 and L3; input: 0.00074 g L⁻¹; hereafter L-mesocosms) and three high slurry concentration mesocosms (H1, H2 and H3; input: 0.0067 g L⁻¹; hereafter H-mesocosms) (Basso et al., 2024). The addition doses used in this experiment were chosen based on the results of a previous similar mesocosm experiment that took place at the ECIMAT-UVIGO facility (Vigo, Spain) in 2022 in which higher doses were added and resulted in important mineral precipitation and flocculation (Basso et al., 2023, 2024). To maintain inorganic nutrients concentrations to levels similar to the initial ones and to ensure repletion throughout the experiment, nutrients were added to all mesocosms on specific days, namely D₅ (ammonium and nitrate) and D₇ (phosphate).

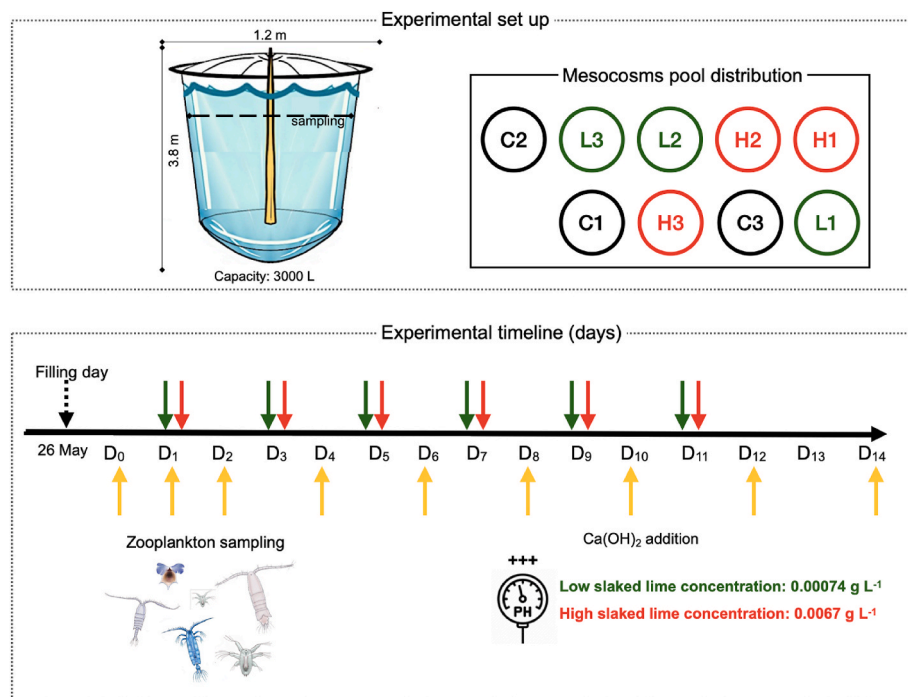


Fig. 1. Schematic representation of the structure of the mesocosms, the replicate treatments and the timeline of the experiment activities. The slaked lime additions are shown in green and red arrows according to the concentration and the sampling for zooplankton is highlighted by yellow arrows. Treatments were as follows: C: control, L: low concentration slaked lime, H: high concentration slaked lime. Illustration credits C. Traboni. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

2.2. Zooplankton sampling and processing

The zooplankton sampling for estimating abundance and diversity took place every other day (Fig. 1). Water from all 9 mesocosms was collected in 10 L carboys using a syringe and a silicon tube connected to the internal environment of each mesocosm. Subsequently, the carboys were transported to the laboratory, and the water was gently siphoned out through silicon tubes, then sieved through 45 µm mesh. The organisms were concentrated in 120 mL plastic vials with a final volume of 30 mL, and they were spiked with 70 % Ethanol to ensure instantaneous preservation of the samples. After gentle and thorough mixing, subsamples of 15 mL were placed on previously gridded glass Petri dishes or Bogorov counting chambers with a 5 mL pipette and zooplankton were counted under the dissecting microscope (Zeiss Stemi 305, magnification 4x). If the subsamples did not contain enough organisms (<100 individuals), the entire sample was counted. Organisms were classified by taxonomic groups to the phylum and class level and for specific groups, also to the order level. For copepods, in particular, the classification included developmental stages (nauplii, copepodites, adults) when possible. Only for the copepods, prosome length across ontogeny (nauplii, copepodites and adults) was recorded through microscope analyses, yet without differentiating between taxonomic orders (Calanoida, Cyclopoida, Harpacticoida; only on D₈). After counting, samples were preserved with formaldehyde (2 % final concentration) and stored at 4 °C.

2.3. Data analysis and statistics

We calculated the Shannon-Weaver index (H') (Shannon and Weaver, 1949) based on the taxa composition and their abundances to estimate the community diversity for the different treatments (Control, Low and High) and the experimental days.

To assess the effect of Ca(OH)_2 addition on zooplankton abundance over time, we employed a linear regression model (LM), which accounts for repeated measures and the hierarchical structure of the data. The

analysis was conducted in the statistical software R (R Core Team, 2019) using the *stats* package (Chambers, 1992). The response variable was zooplankton abundance, while the fixed factors included treatments (Control, Low and High) and time (measurements every other day). The LM was specified as follows:

$$\text{Abundance}_{ij} = \beta_0 + \beta_1 (\text{treatment}_i) + \beta_2 (\text{time}_j) + \beta_3 (\text{treatment}_i \times \text{time}_j) + \varepsilon_{ij}$$

where β_0 is the intercept, β_1 , β_2 , and β_3 represent the fixed effects of treatment, time, and their interaction, respectively. The term ε_{ij} accounts for the residual error for each observation.

The model included the interaction between treatment and time to determine whether the effect of Ca(OH)_2 addition on zooplankton total abundance varied over the course of the experiment. The homogeneity of variances for the zooplankton abundance across treatments (Control, Low and High) was tested using Levene's test with *car* package. If significant, a ANOVA Welch test was carried out. Subsequently, we performed *post hoc* pairwise comparisons to identify specific group differences at each time point, using Tamhane's test. Same procedure of the LM was employed to evaluate copepod abundance evolution, differentiated by stage (nauplii, copepodites and adults).

Redundancy Analysis (RDA, after Legendre and Legendre, 2012) was used to investigate the relationship between zooplankton community structure and environmental variables, namely pH, temperature, nitrate, ammonium, and chlorophyll *a*, and treatments (Control, Low and High). Additionally, we conducted Pearson correlation analyses to estimate the strength and direction of the relationships between zooplankton abundance and environmental parameters. In the case of the phosphate concentration was used for correlation and not for RDA, as some samples were contaminated.

All calculations were performed using the *stats* package in R Statistical Software (R Core Team, 2019). All variables expressed in absolute value are reported as mean values $\pm$ standard deviations. Statistically significant differences were considered at $p < 0.05$ level.

3. Results

3.1. Zooplankton abundance, community composition and diversity

The overall average of zooplankton abundance at the beginning of the experiment was 1132 ± 361 ind L^{-1} , considering all 9 mesocosms (D_0). In control conditions (C-mesocosms), the abundance gradually increased to reach a peak on D_{10} (4700 ± 680 ind L^{-1}), followed by a rapid decline by the end of the experiment on D_{14} (1937 ± 218 ind L^{-1} ; Fig. 2). In mesocosms treated with low $Ca(OH)_2$ concentration (L-mesocosms), the increase in D_{10} was not as accentuated as in the control conditions (2443 ± 802 ind L^{-1}), but the final zooplankton abundance recorded in these mesocosms was similar (1953 ± 330 ind L^{-1}) to the one in C-mesocosms. The H-mesocosms were, instead, characterized by a rapid zooplankton abundance decrease recorded on D_6 (292 ± 242 ind L^{-1}), followed by a rapid recovery peak on D_{12} (3753 ± 258 ind L^{-1}), ultimately descending to similar values as in the other two conditions (1495 ± 351 ind L^{-1}) by the end of the experiment on D_{14} (Fig. 2).

The taxonomic categories found were the following: Radiolaria, Annelida, Bryozoa (larvae), Ciliophora (Tintinnina), Cirripedia (larvae), Copepoda (Calanoida, Cyclopoida, Harpacticoida), Mollusca (Bivalvia, Gastropoda), Rotifera. Individuals representing a low percentage (less than 1 %) within the community composition were grouped under the name of "Others" (Annex I). The most abundant taxa were represented by Copepoda and Ciliophora (Fig. 3); from the phylum Ciliophora only tintinnids were counted, hence for simplicity we refer to them as simply Tintinnina.

A similar trend was observed in C-, L-, and H-mesocosms. The initial community was characterized by mostly tintinnins outnumbering any other group from day D_1 to D_4 . However, on D_6 the zooplankton community experienced a transition phase where a mixed and more diverse assemblage was established (Fig. 3), including tintinnins, copepods, mollusks, annelids and radiolarians. From D_8 until the end of the experiment, the zooplankton community was almost totally dominated by copepods. The relative contribution to the community was the highest at the beginning for tintinnids (96 % on D_2), and at the end of the experiment (100 % on D_{12}) for copepods; the other less abundant groups were present in detectable abundance only during the central days (D_4 – D_8) of the experiment, and they accounted for 4–23 % of the community, represented by radiolarians, annelids, and mollusks. It is important to highlight that as samples were sieved (45 μ m), therefore smaller tintinnids as well as nauplii could be underestimated.

Among the identified groups, unicellular zooplankton, specifically

tintinnids and radiolarians, were annotated (Fig. 3). Radiolarians occurred in very low numbers throughout the experiment, and their abundance showed an overall and similar decreasing trend over time in C- and H-mesocosms, but a transient peak was recorded on D_4 in L-mesocosms (Fig. 3). Annelids and mollusks peaked on D_6 in two treatments (C- and L-mesocosms; Fig. 3), whereas in H-mesocosms, a peak was not evident for either taxon. The abundance of both metazoan groups in H-mesocosms was characterized by a progressively decreasing trend and an overall lower abundance of both annelids and mollusks.

The highest taxa diversity index was observed in C-, L-, and H-mesocosms on D_6 , but control and L-mesocosms presented higher values ($H' = 2.1$ in both cases) compared to those found in H-mesocosms ($H' = 1.6$; Fig. 4). After this day, the diversity index decreased for all treatments and finally showed a slight recovery, with values increasing on D_{14} ($H' = 0.86$, 0.73 , and 0.8 , for C-, L-, and H-mesocosms, respectively). In contrast, the lowest diversity index for the zooplankton community was found on D_2 for the L and H-mesocosms ($H' = 0.33$ and 0.43 , respectively), followed by D_{10} for the H-mesocosms ($H' = 0.5$). The diversity index started with similar values for the three treatments, showing only a decrease for the L-mesocosms at the end of the experiment.

The LM revealed that experimental time (Day) had a significant effect on zooplankton abundance, with abundance increasing significantly as the experiment progressed ($p = 0.001$; Table 1). However, the treatment effects of both L- and H-mesocosms, as well as their interactions with time, were not statistically significant ($p = 0.12$ and 0.53 for L- and H-mesocosms, respectively), indicating that the additions did not have a measurable impact on zooplankton abundance when compared to the control group over the experimental period. The zooplankton abundance differences of treatments at each time point were statistically significant in the following days: D_6 H-mesocosms were different from C- and L-mesocosms ($p < 0.05$, respectively), as well as for D_{10} when C-, H- and L-mesocosms presented differences ($p < 0.05$) with control abundance being higher than H-mesocosms and much higher than L-mesocosms ($p < 0.001$), and at D_{12} there are still significant differences between H- and the C- and L-mesocosms ($p < 0.01$).

3.2. Relationships between zooplankton abundance and environmental factors

RDA ordination diagram (Fig. 5) showed that the relationship between zooplankton community structure and environmental variables, namely pH, temperature, nitrate, ammonium, and Chla, and treatment (Control, Low and High) explained 37.6 % of the total variance in the zooplankton community abundance. The first two RDA axes explained 35.4 % and 2.2 % of the total variance, respectively, indicating that the majority of the zooplankton-environment relationship was captured by the first axis. The RDA model was statistically significant ($F = 8.16$, $p < 0.001$), suggesting that the environmental variables collectively exerted a strong influence on zooplankton groups. However, treatment effects were not strongly aligned with the main RDA axes, indicating a limited direct influence compared to the environmental variables. The variance partitioning revealed that constrained components accounted for a substantial proportion of the variance (0.12), further emphasizing the influence of environmental gradients and the minimal contribution of treatments to the overall explained variance. The biplot indicated that pH and nutrient concentrations (nitrate and ammonium, NO_3 and NH_4 , respectively in Fig. 5) were closely aligned with the first RDA axis, suggesting that these factors might be primary drivers of zooplankton community structure in the study. Chlorophyll *a* was more weakly associated with the second RDA axis.

Analysis of the species scores on the first RDA axis (RDA1) indicated that Copepoda and Tintinnina were the most strongly influenced groups, with Copepoda being positively associated with the environmental variables (pH and temperature) and Tintinnina showing a strong negative association with them. Other taxa, such as Rotifera, Radiolaria, Annelida, and Mollusca exhibited weaker relationships with the

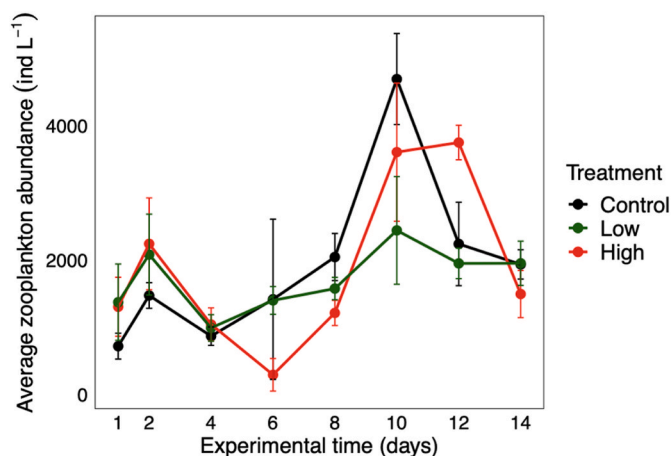


Fig. 2. Zooplankton total abundance (ind L^{-1}) over experimental time for three different treatments: Control (black), Low (green), and High (red) mesocosms. Points represent daily measurements, mean $\pm$ SD of three replicates. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

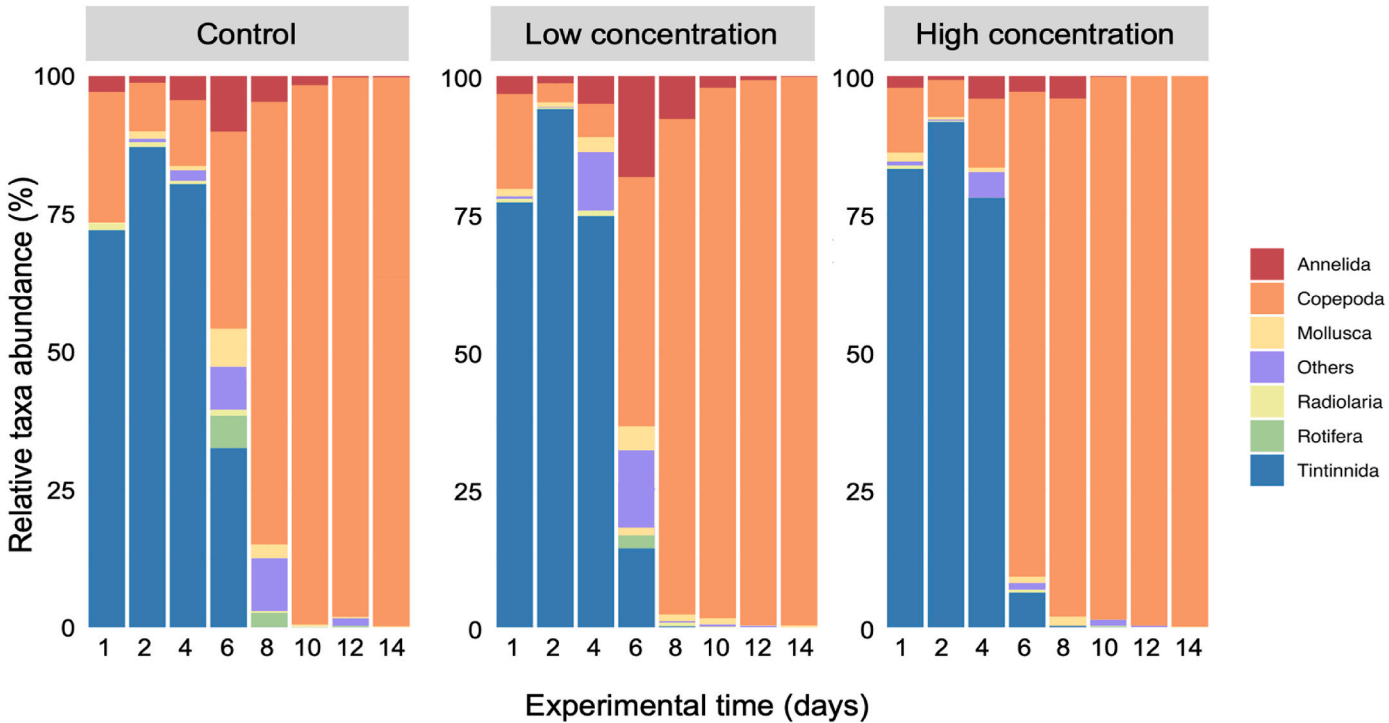


Fig. 3. Zooplankton community composition recorded in the three treatments (from left to right Control, Low and High) throughout the duration of the mesocosm experiment (see Fig. 1).

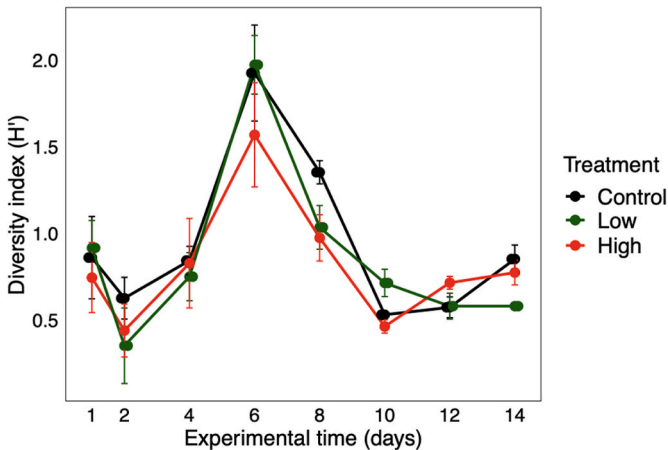


Fig. 4. Shannon diversity index (H') $\pm$ standard deviation for the zooplankton community abundance throughout the duration of the mesocosm experiment for the three treatments (Control, Low and High).

Table 1
Summary of the linear regression model examining the relationship between zooplankton abundance, treatments, day, and their interaction.

	Value	Std Error	df	t-value	p-value
(Intercept)	853.05488	384.80330	60	2.2168596	0.030
Treatment High	236.67474	544.19405	6	0.4349087	0.679
Treatment Low	519.21711	544.19405	6	0.9541029	0.377
Day	150.43382	45.95178	60	3.2737320	0.001
Treatment High x Day	-41.03914	64.98563	60	-0.6315110	0.530
Treatment Low x Day	-101.42521	64.98563	60	-1.5607328	0.123

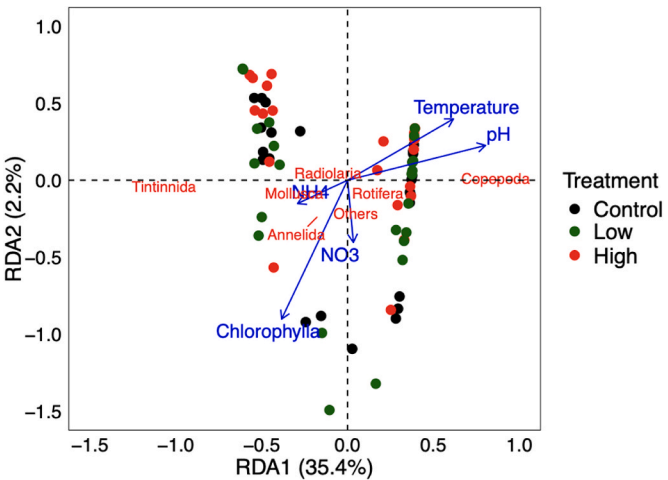


Fig. 5. Redundancy analysis (RDA) ordination of zooplankton groups abundance, environmental variables (temperature, chlorophyll *a* and nutrients) and treatments (Control, Low and High) during the experimental period. NO₃: nitrate; NH₄: ammonium.

environmental gradients along RDA axes. Therefore, environmental factors, particularly pH, temperature and nutrients, significantly shaped the zooplankton community, with Copepoda and Tintinnina being the most responsive taxa to these gradients. The high percentage of variance explained by RDA1 highlights the dominant role of these variables in structuring the zooplankton assemblage.

Pearson correlations reveal relationships between the seven zooplankton groups and some environmental variables (Fig. 6). Copepoda presented a significant positive correlation with pH and temperature, indicating that the abundance increases with higher levels of these parameters. In contrast, Copepoda abundance showed a significant negative correlation with phosphate (PO₄, Fig. 6), Chl*a* and ammonium

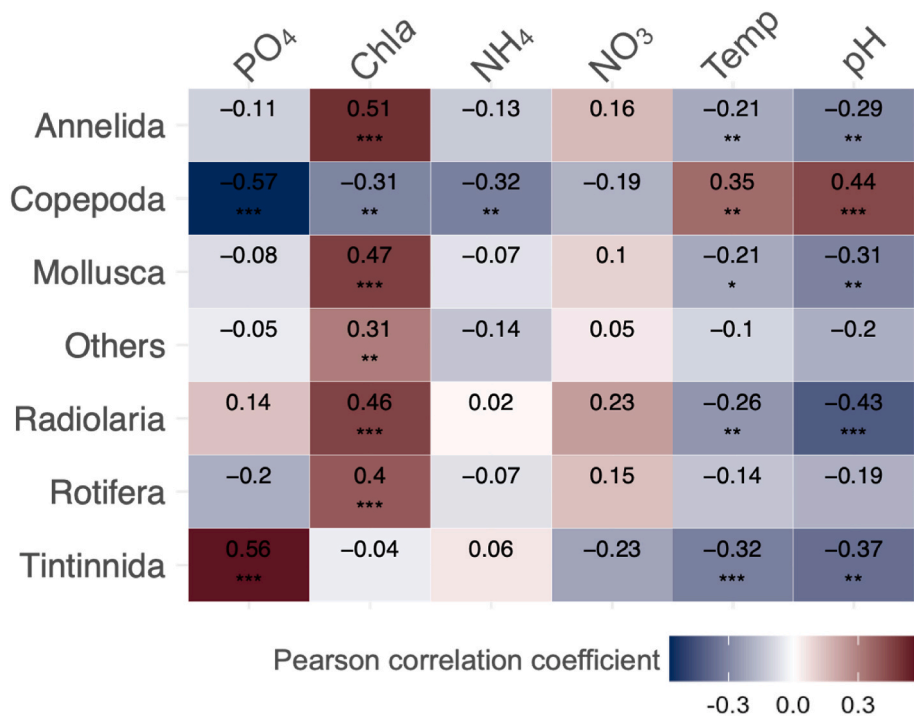


Fig. 6. Heatmap of Pearson correlations showing the strength and direction of relationships between zooplankton group abundances and environmental variables such as pH, temperature (Temp), nutrients (phosphate, nitrate and ammonium, PO₄, NO₃ and NH₄, respectively), and chlorophyll *a* (Chla). Significant correlations are marked with asterisks (*: *p* < 0.05, **: *p* < 0.01 and ***: *p* < 0.001, respectively).

(NH₄, Fig. 6). Tintinnida exhibited a significant positive correlation with phosphate, suggesting that the abundance increases with higher concentrations of this nutrient, while negative relationships were found for temperature and pH. Similarly, the remaining groups, namely Annelida, Mollusca, Radiolaria, Rotifera and others, were positively correlated with Chla, but negative significant correlations were found with temperature and pH, except for Rotifera and others groups.

3.3. Copepod ontogeny and taxonomic groups composition

Nauplii were the most abundant group (10–4680 ind L⁻¹),

characterized by low abundances at the beginning of the experiment (133 ± 29 ind L⁻¹ considering all mesocosms) that rapidly increased to reach a maximum on D₁₀ (3148 ± 1045 ind L⁻¹; Fig. 7). On this day, the highest abundance was registered for the C-mesocosms (4147 ± 586 ind L⁻¹), followed by the H-mesocosms (3233 ± 865 ind L⁻¹), and exhibiting the lowest abundance in the L-mesocosms (2063 ± 686 ind L⁻¹). After D₁₀, nauplii decreased in number until the end of the experiment in all treatments (1156 ± 232 ind L⁻¹ on average on D₁₄).

Copepodites started to develop by D₆, reaching their highest abundance at the termination of the experiment on D₁₄ in the case of the C and L-mesocosms (635 ± 139 and 588 ± 131 ind L⁻¹, respectively;

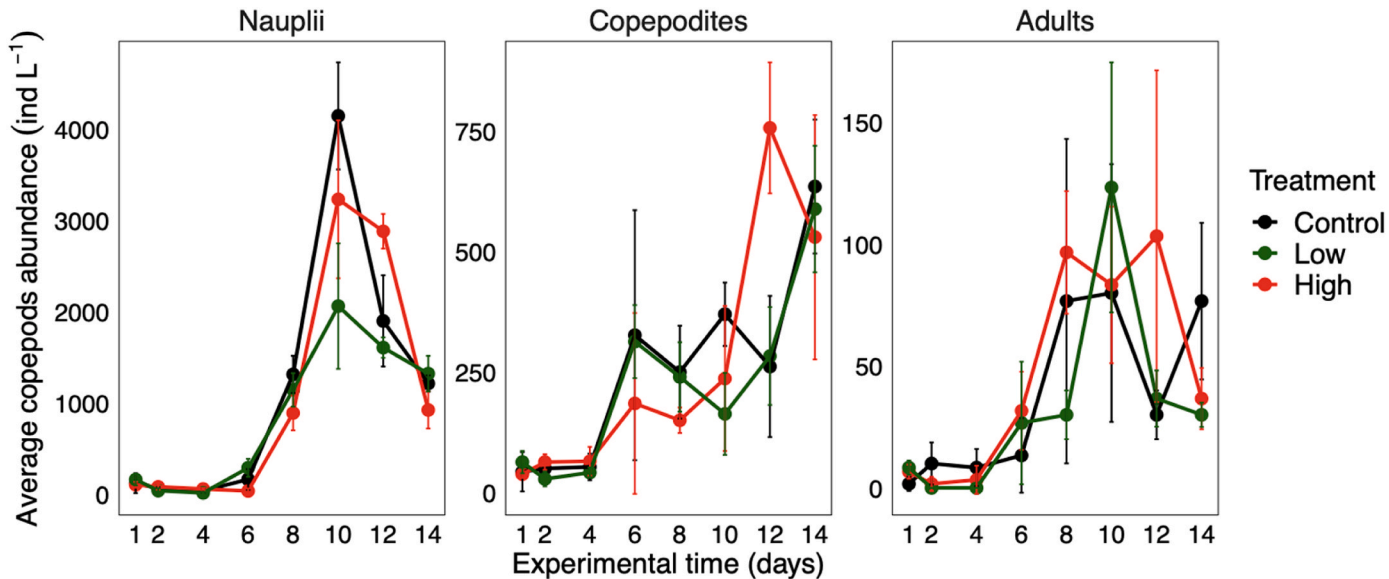


Fig. 7. Average copepods abundance (ind L⁻¹) ± standard deviation recorded for the three mesocosm treatment groups over the duration of the experiment, differentiated by their ontogeny stage (nauplii, copepodites and adults from top to bottom panels).

Fig. 7), while for H-mesocosms the highest abundance was registered on D₁₂ (757 ± 136 ind L⁻¹) decreasing by D₁₄ but being similar in numbers to the other treatments (530 ± 253 ind L⁻¹). The copepodites followed a similar pattern across all treatments, with very similar values observed between the C- and L-mesocosms, while the H-mesocosms exhibited a delayed peak, reaching the highest numbers on D₁₂, rather than D₁₀ like the other treatments.

Adult copepods attained their maximum average abundance on D₁₀, in the C- and L-mesocosms (80 ± 53 and 123 ± 51 ind L⁻¹, respectively), while for H-mesocosms, the peak was recorded on D₁₂ (103 ± 68 ind L⁻¹). Although all treatments exhibited the same general pattern of copepod growth, with increases in copepodite and adult stages after D₆, the nauplii phase showed clear differences in peak abundances. The C-mesocosms exhibited the highest peak abundance, followed by the H-mesocosms and L-mesocosms. Overall, the abundance of naupliar stage exceeded that of the copepodites by an order of magnitude, which in turn was an order of magnitude higher than the abundance found for adult copepods, with the C-mesocosms consistently showing the highest numbers at each stage.

The LME model result did not reveal significant differences between treatments, day and each copepod stages (nauplii, copepodites and adults), except for the nauplii and day interaction, which indicates that nauplii abundances increased significantly over time ($p < 0.001$; Table 2) as the development progressed during the experiment. In each treatment (Control, Low and High), significant differences were found between nauplii and copepodite abundances ($p < 0.001$ in all cases), as well as between abundances of nauplii and adults ($p < 0.001$ in all cases). In fact, in some cases, nauplii were significantly more abundant (up to one order of magnitude higher), compared to copepodites or adults.

Concerning copepod body size, it was found that the average prosome length was 118 ± 35 μ m for nauplii, 420 ± 122 μ m at the copepodite stage, and 702 ± 207 μ m for adults (measured only on D₈, data not shown).

When taxonomic orders were taken into account for copepodites and adult copepods (Fig. 8), calanoids were the most abundant, especially

Table 2

Summary of the linear model examining the relationship between copepod stages abundance (nauplii, copepodites and adults), treatments, day, and their interaction. StageC: copepodites. StageN: nauplii.

	Value	Std Error	df	t-value	p-value
(Intercept)	-2.8450	203.646	198	-0.014	0.989
StageC	-18.3575	287.999	198	-0.064	0.949
StageN	-220.447	287.999	198	-0.765	0.445
Treatment High	2.9661	287.999	198	0.010	0.992
Treatment Low	3.2163	287.999	198	0.011	0.991
Day	5.6040	24.3187	198	0.230	0.818
StageC x Treatment	-58.6562	407.292	198	-0.144	0.886
High					
StageN x Treatment	-92.5706	407.292	198	-0.227	0.820
High					
StageC x Treatment	-0.9605	407.292	198	-0.002	0.998
Low					
StageN x Treatment	28.5109	407.292	198	0.070	0.944
Low					
StageC x Day	32.2841	34.3918	198	0.939	0.349
StageN x Day	183.367	34.3918	198	5.332	$2.6 \times 10^{-7*}$
Treatment High x Day	0.7533	34.3918	198	0.022	0.983
Treatment Low x Day	-1.1824	34.3918	198	-0.034	0.973
StageC x Treatment	7.6769	48.6374	198	0.158	0.875
High x Day					
StageN x Treatment	-1.3936	48.6374	198	-0.029	0.977
High x Day					
StageC x Treatment	-3.8418	48.6374	198	-0.079	0.937
Low x Day					
StageN x Treatment	-43.9723	48.6374	198	-0.904	0.367
Low x Day					

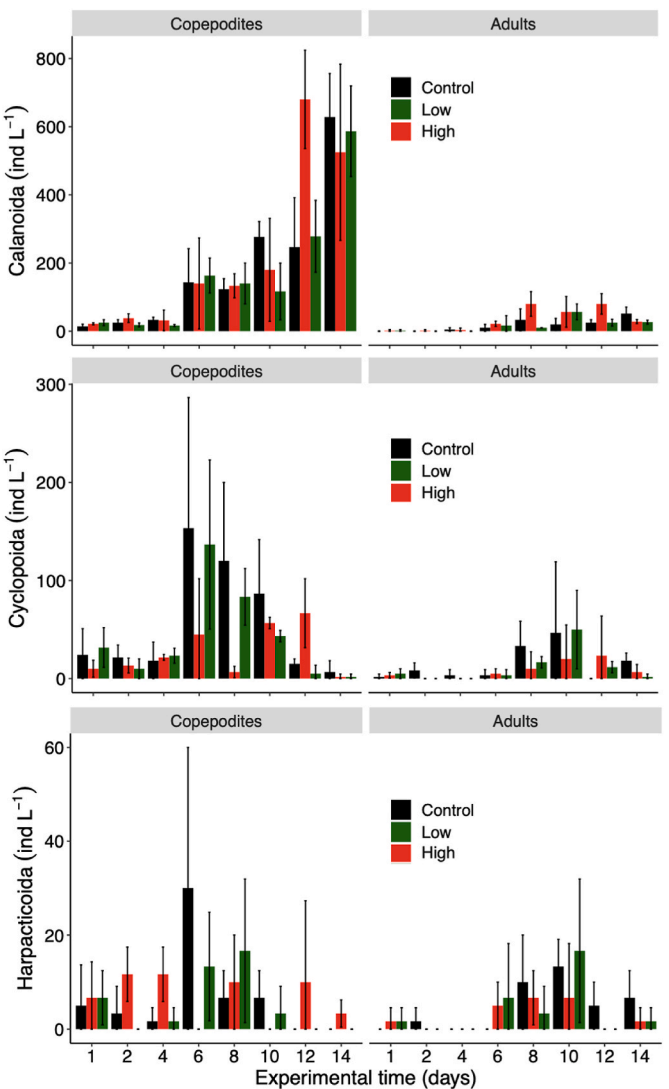


Fig. 8. Average abundance of adults and copepodites stages of copepods (ind L⁻¹) $\pm$ standard deviation recorded in the three treatments over the duration of the experiment. Copepods are classified according to the three main taxonomic orders Calanoida, Cyclopoida and Harpacticoida.

copepodites, followed by cyclopoid copepodites, whereas harpacticoids were 10-fold less abundant than the former. The abundance of adults on average was very low in all cases compared to copepodites (12 ± 13 and 80 ± 50 ind L⁻¹, respectively).

4. Discussion

Our study aimed to assess the response to ocean alkalinity enhancement (OAE) of a coastal zooplankton community in the oligotrophic eastern Mediterranean by testing two different slaked lime (Ca (OH)₂) concentrations. Over the past decades, ocean acidification has been at the center of discussions within the research community (Tilbrook et al., 2019). Now, interventions such as OAE are becoming extremely important given the urgent need to identify remedial strategies to counteract the effects of ocean acidification, while also enhancing the ocean's capacity for carbon sequestration. Ocean liming, a key method within OAE, not only helps mitigate acidification by increasing the ocean's buffering capacity but also accelerates the natural processes of CO₂ removal (Feng et al., 2017; Foteinis et al., 2022; Hartmann et al., 2022; Suitner et al., 2024), turning atmospheric carbon into stable, long-term forms such as bicarbonates. As a result, these

strategies hold promise not only for protecting marine ecosystems but also for contributing to global efforts to reduce atmospheric CO₂ levels. Here, we found similar zooplankton community compositions between the control and the two slaked lime treatments, but taxa diversity demonstrated a lower index for the community in the H-mesocosms. Total abundances for zooplankton organisms were similar between control and low Ca(OH)₂ concentration mesocosms, but were slightly less abundant in H-mesocosms, this difference being significant between L- and H-mesocosms on four occasions (D₆, D₈, D₁₀ and D₁₂). Tintinnids exhibited the highest abundances at the beginning of the experiment, but were later replaced by copepods, with the Calanoida order being the most abundant group. Abundant Copepoda nauplii appeared first; later the number of copepodites increased and adults showed higher abundances in the middle of the experimental period.

We did not observe significant effects of low or high Ca(OH)₂ additions on total zooplankton abundance compared to the control throughout the experimental period, although some differences in abundance were noticeable. These findings contrast with some previous studies, which suggest that OAE could alter marine plankton composition, affecting biogeochemical processes such as nutrient cycling, primary production, and carbon sequestration (Paul et al., 2024). Furthermore, other research indicates that changes in zooplankton abundance may vary across different groups in response to OAE (Guo et al., 2024). In this context, while we did not find significant overall effects, our results highlight the potential for some zooplankton groups to be impacted by higher concentrations of Ca(OH)₂, a trend observed in other studies. We present greater details about our results below.

4.1. Zooplankton community succession

The peak in total zooplankton abundance observed on day D₁₀ may be due to the absence of grazing on this group and a delayed response to nutrient additions (D_{5,7}), which boosted phytoplankton growth; nutrients were subsequently transferred via the grazing chain towards higher trophic levels. The decline in total abundance towards the end of the experiment could be attributed to two factors, either acting independently or in combination: food availability limitation along with an increase in heterotrophic organisms (replacement of naupliar copepods by copepodites and adults, as discussed further in the text), coupled with the settlement of meroplanktonic organisms that complete their pelagic life stage (e.g., gastropods and annelids) present in the zooplankton community of this marine system (this study; Zervoudaki et al., 2020). This was also confirmed by the decrease in zooplankton diversity indexes recorded by the end of the experiment. The results suggested a generally homogeneous growth pattern across treatments, with some occasional differences that remain unexplained, but which returned to the same level of homogeneity by D₁₄. Significant differences in total zooplankton abundance were found two times between H-mesocosm vs. C- and L-mesocosms (D₆ and D₁₂), and once between C- and L-mesocosms (D₁₀), while Chl_a concentrations (as a proxy of phytoplankton biomass) were similar on those days (Serret et al., 2024; José González Fernández, personnel communications). We hypothesize that ineffective prey selection may have occurred due to the persistent presence of suspended particles released from the addition of Ca(OH)₂, particularly in the L-concentration mesocosms, resulted in lower abundances recorded on D₁₀, while for the other three days, an indirect effect on prey presence in H-concentration mesocosms could affect the zooplankton growth. In contrast, in the H-concentration mesocosms, precipitation and sedimentation of these particles seemed more pronounced, accumulating on the bottom (Basso et al., 2024).

Copepoda and Ciliophora (Tintinnina) were the most abundant groups in the seawater community enclosed by the mesocosms throughout the experiment. The community transitioned from being tintinnids-dominated (96 % of the community on D₂) to being almost entirely copepod-dominated by the end. Species composition, abundance and biomass of Tintinnina have been associated with

environmental factors, such as nutrients, pH, and salinity in the south-eastern Mediterranean (Heneash et al., 2015), as well as with phytoplankton size diversity along the west-to-east longitudinal transect in the Mediterranean Sea (Dolan et al., 2002). In our study, additions of slaked lime and pH were weakly and negatively correlated with Tintinnid abundance, in contrast to nutrients, specifically phosphate. This is the first study to indicate a decrease in tintinnids abundance related to pH, although other parameters may also be interacting with it. Species composition, abundance and biomass of tintinnids have been associated with environmental factors, for example, as we observed the progressive increase in phosphate concentration was concomitant with the increase in tintinnid population in agreement with Pitta et al. (2024). In contrast, we did not find a significant correlation with nitrate, while in a study conducted in the Arabian Sea, tintinnids were found highly correlated with environmental nitrate concentration (Dorham and Abdel-Aziz, 2001). Tintinnids are also considered key primary consumers (Pitta et al., 2001; Calbet and Landry, 2004), playing an essential role in the food web (Pierce and Turner, 1992) in oligotrophic waters, where pico- and nanoplankton dominate. Our results showed a significant positive relationship between Tintinnida and phosphate, as they typically respond to phosphate enrichment with increased food availability, as previously reported by Pitta et al. (2016). In a parallel dilution experiment, conducted under the same experimental conditions, intermediate-sized aloricate ciliates were observed (15–35 µm; Traboni et al., under review), primarily utilizing mixotrophic and heterotrophic feeding strategies. The abundance curves for copepod nauplii and tintinnids reveal an important decline in tintinnid populations during the initial days, followed by a marked increase in nauplii. This pattern reflects a classic predator-prey oscillation, where the nauplii heavily graze on tintinnids, resulting in a substantial increase in nauplii populations and a sharp decrease in tintinnids. Previous works have established that copepods prefer ciliates over autotrophic preys (Wiadnyana et al., 1989; Calbet and Saiz, 2005) with clearance rates being one order of magnitude higher on ciliates than on chlorophyll *a* and providing adequate quality and size food supply compared to primary producers (Zervoudaki et al., 2007). At the same time, the predation pressure on *Synechococcus* and pigmented flagellates decreased (Traboni et al., under review); these two groups are generally considered prey for microzooplankton grazers. The observed shift in tintinnid abundances may have multiple impacts on energy transfer, as well as nutrient and carbon cycling in coastal waters (Biswas et al., 2011).

4.2. Copepods dynamics under ocean liming

An important observation in this study relates to the dynamics of the copepod nauplii populations. Initially, the abundance of nauplii was low; however, their numbers exhibited a rapid increase throughout the duration of the experiment. This finding suggests that the applied treatments did not have a significant negative impact on the hatching success of copepod eggs (Annex II). However, it is important to note that the abundance of nauplii varied across the different treatment groups. This variability could provide valuable insights into how the treatments differentially affect nauplii development and recruitment, and, consequently, their population fitness. Furthermore, the timing of critical life cycle events, such as egg production, hatching, and the development of various life stages, can vary in response to seasonal changes in temperature and nutrient availability. Variations in copepod populations may have significant implications for the marine food web, affecting survival, disrupting predator-prey dynamics, affecting energy transfer and ecosystem stability (Costanza et al., 1997; Turner, 2015).

Copepods transfer biomass from primary producers to higher trophic levels and, by doing so, they contribute significantly to the vertical particle flux, influencing global biogeochemical cycles (Mauchline, 1998; Turner, 2015; Steinberg and Landry, 2017). They present complex life cycles, passing through different stages during their ontogeny development (Atkinson, 1998), which make effects and predictions

more difficult to elucidate. We did not observe any effects of the different levels of $\text{Ca}(\text{OH})_2$ treatments across the different copepod life stages analyzed during the experiment. Contrary, [Camatti et al. \(2024\)](#) found that, even for the notably tolerant species *Acartia tonsa*, alkalization had negative effects (on copepod mobility and survival) only after prolonged exposure (>6 h) at pH 9 or higher tested during a short period of time. In this line, [Pedersen and Hansen \(2003\)](#) also found that copepods are quite sensitive in terms of grazing abilities to high pH levels compared to other planktonic organisms. In a recent study, the indirect effects of OAE on the copepod *Temora longicornis* were assessed. While prey growth was negatively affected, its nutritional quality improved, potentially compensating for reduced prey availability and higher respiration rates recorded, thus enhancing the copepod's resilience to OAE-induced changes in pH ([Bhaumik et al., 2025](#)). While only a few studies have focused on OAE effects on copepods, much more is known about the effects of acidification and warming on these organisms, impacting on their performance, egg hatching and rates of survival mostly in nauplii and copepodites development stage (e.g., [Weydmann et al., 2012](#); [Pedersen et al., 2013](#); [Mayor et al., 2015](#)). On the contrary, the positive correlations found between the Copepoda group, temperature and pH differ from those reported in the mentioned studies, without clarifying whether these parameters act independently or additively on their abundance.

Our results align with the observation that calanoid copepods account for approximately 80 % of the global mesozooplankton abundance and biomass in marine pelagic food webs ([Longhurst, 1985](#); [Mauchline, 1998](#)). During the experiment Calanoida abundance was followed by Cyclopoida, mostly represented by copepodite stages compared to adults. Carnivorous species in the mesocosms, such as *Oithona* spp. (the main Cyclopoida observed), may have significantly preyed on micro-zooplankton, potentially influencing phytoplankton abundance through trophic cascades ([Calbet et al., 2012](#)), favoring more herbivorous copepod groups or species belonging to the Calanoida class, such as *Ctenocalanus* spp. and *Paracalanus* spp. identified during the experiment. Although no significant differences were found between treatments for the different orders of copepods, it is important that future studies consider the different feeding habits and life strategies of these groups to better establish the potential implications of the proposed alkalization method. Copepods exhibit a range of feeding strategies and some species are strict filter feeders, relying on suspended particles, such as phytoplankton or detritus, while others may actively hunt smaller zooplankton or even engage in omnivorous feeding ([Durbin et al., 1990](#); [Durbin et al., 1995](#); [Castellani et al., 2008](#); [Lee et al., 2012](#)). The lack of adaptation to environmental changes caused by AOE could favor certain groups over others, potentially reducing diversity or allowing only specific groups to proliferate. Recent research by [Bhaumik et al. \(2025\)](#) has found changes in ingestion rates with increasing alkalinity, emphasizing the need for deeper understanding of the implications for copepod metabolic processes, as well as their relationship with prey abundance and nutritional quality.

Other taxa, such as Annelida, Mollusca, and Radiolaria, were present but in much lower numbers, with less pronounced changes in abundance and/or composition under the applied treatments. The negative relationship observed between their abundances and pH suggests that more attention should be given to these groups, particularly in ecosystems where their presence is more abundant. Despite being a group known for its high presence in the eastern Mediterranean Sea ([Hannides et al., 2015](#); [Zervoudaki et al., 2020](#) and references therein), no individuals from the order Diplostraca (formerly Cladocera) were recorded in this study ([Siokou et al., 2014](#)). Mesocosms are especially valuable for studying and manipulating ecosystems under controlled, replicated, and repeatable conditions. Future mesocosm studies should account for direct and indirect effects on zooplankton communities overall the food web of liming addition. The timing of liming addition events may be a critical factor in determining the impact on zooplankton communities, as different physiological states and complex life cycle stages can lead to

varying effects. This study primarily focused on changes in abundance and community composition, but future research should explore the potential impacts of OAE on reproduction, as well as possible anatomical or physiological effects that may not occur as quickly but could have significant consequences for offspring.

5. Conclusion

During this study, we observed that zooplankton total abundance was not significantly affected by the addition of $\text{Ca}(\text{OH})_2$. The treatment effects suggest that $\text{Ca}(\text{OH})_2$ additions, even at varying concentrations, did not significantly influence overall zooplankton abundance or diversity within the limits of the lime concentration gradient we tested, except on specific days. Correlations indicated that the abundance of specific groups, particularly Tintinnina and Copepoda, was primarily driven by environmental factors such as pH, temperature, and nutrient concentrations. These environmental variables shaped the zooplankton community composition and dynamics. It is equally important to highlight that this study was conducted over a short time period in a known oligotrophic marine coastal system, specifically the eastern Mediterranean, and the effects observed in this particular ecosystem may differ in other regions, potentially enhancing and/or attenuating other parameters.

Under a future approach of OAE by $\text{Ca}(\text{OH})_2$ additions in the Eastern Mediterranean Sea, major impacts on the zooplankton community structure are not expected. However, it is possible that treatments affecting copepod populations could influence the efficiency of energy transfer to higher trophic levels. Copepods play a crucial role in the marine food web, but if their ability to efficiently capture prey or store lipids is impaired by changes in their environment, this could lead to a reduced energy transfer to predators such as fish and larger zooplankton. A decrease in the nutritional quality of copepods, in terms of lipid content, could impact higher trophic levels, which rely on these zooplankton as a primary food source. Therefore, the results should not be indiscriminately extrapolated and must be interpreted with caution. Other marine ecosystems may respond very differently to alterations in their chemistry through ocean liming. For example, environments with a different plankton composition, or systems with non-oligotrophic waters, may exhibit different responses. In nutrient-poor systems, the effects on plankton community structure and energy transfer could be more pronounced due to the more limited availability of resources and slower metabolic rates of the organisms.

In addition, ocean alkalization involves a significant global-scale geo-engineering challenge and therefore cannot be pursued responsibly without thorough research, monitoring, governance, and regulation. Future research should explore the long-term effects of altered pH and nutrient conditions on zooplankton communities, focusing on species-specific and life-stage responses to environmental stressors. Integrating additional ecological factors, such as predation pressure and food availability briefly addressed in parallel to this study ([Traboni et al., under review](#)), alongside molecular approaches to assess functional diversity, could provide deeper insights into the ecological roles of zooplankton under changing environmental conditions.

CRedit authorship contribution statement

Ariadna C. Nocera: Writing – review & editing, Writing – original draft, Formal analysis, Data curation, Conceptualization. **Claudia Traboni:** Writing – review & editing, Writing – original draft, Data curation. **Justine Courboulès:** Writing – review & editing, Writing – original draft, Data curation. **Filomena Romano:** Writing – review & editing, Writing – original draft, Data curation. **Christos Chantzaras:** Writing – review & editing. **Iordanis Magiopoulos:** Writing – review & editing, Conceptualization. **Selene Varliero:** Writing – review & editing, Data curation. **Arianna Azzellino:** Writing – review & editing, Formal analysis. **Pietro Bazzicalupo:** Writing – review & editing. **Piero**

Macchi: Writing – review & editing. **Guido Raos:** Writing – review & editing. **Daniela Basso:** Writing – review & editing, Project administration, Funding acquisition, Conceptualization. **Paraskevi Pitta:** Writing – review & editing, Writing – original draft, Funding acquisition, Conceptualization.

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Annex I.

List of zooplankton groups identified in the samples during the experimental period, without considering the treatments and specific sampling days. Classification according to [Swadling et al. \(2013\)](#).

Life strategy - Phylum - Class - Order
Holoplankton
Ciliophora
Tintinnina
Radiolaria
Foraminifera
Arthropoda
Copepoda
Nauplii
<i>Calanoida</i> (copepodites)
<i>Cyclopoida</i> (copepodites)
<i>Harpacticoida</i> (copepodites)
<i>Calanoida</i> (adults)
<i>Cyclopoida</i> (adults)
<i>Harpacticoida</i> (adults)
Thecostraca
Nauplii
Meroplankton
Mollusca
Trochophora
Veliger
Bivalvia
Gastropoda
Annelida
Bryozoa (larvae)
Rotifera
Unidentified eggs

Annex II.

Abundances of Copepoda stages nauplii (N), copepodites (C), and adults (A), across different days and treatments. Calculated ratios of nauplii to copepodites (N:C ratio) and nauplii to adults (N:A ratio) are present, offering insights into the developmental dynamics within the population.

Day	Treatment	Nauplii abundance	Copepodites abundance	Adults abundance	N:C ratio	N:A ratio
1	C	375	130	5	2.9	75.0
1	H	325	115	20	2.8	16.3
1	L	495	190	25	2.6	19.8
2	C	210	150	30	1.4	7.0
2	H	255	190	5	1.3	51.0
2	L	130	85	0	1.5	–
4	C	125	160	25	0.8	5.0

(continued on next page)

(continued)

Day	Treatment	Nauplii abundance	Copepodites abundance	Adults abundance	N:C ratio	N:A ratio
4	H	185	195	10	0.9	18.5
4	L	55	125	0	0.4	–
6	C	500	980	40	0.5	12.5
6	H	120	555	95	0.2	1.3
6	L	880	940	80	0.9	11.0
8	C	3950	750	230	5.3	17.2
8	H	2680	450	290	6.0	9.2
8	L	3440	720	90	4.8	38.2
10	C	12440	1110	240	11.2	51.8
10	H	9700	710	250	13.7	38.8
10	L	6190	490	370	12.6	16.7
12	C	5705	785	90	7.3	63.4
12	H	8650	2270	310	3.8	27.9
12	L	4830	850	110	5.7	43.9
14	C	3650	1905	230	1.9	15.9
14	H	2780	1590	110	1.7	25.3
14	L	3975	1765	90	2.3	44.2

Data availability

Data will be made available on request.

References

Algueró-Muñiz, M., Alvarez-Fernandez, S., Thor, P., Bach, L.T., Esposito, M., Horn, H.G., Ecker, U., Langer, J.A.F., et al., 2017. Ocean acidification effects on mesozooplankton community development: results from a long-term mesocosm experiment. *PLoS One* 12.

Alvarez-Fernandez, S., Bach, L.T., Taucher, J., Riebesell, U., Sommer, U., Aberle, N., Brussaard, C.P.D., Boersma, M., 2018. Plankton responses to ocean acidification: the role of nutrient limitation. *Prog. Oceanogr.* 165, 11–18.

Atkinson, A., 1998. Life Cycle Strategies of Epipelagic Copepods in the Southern Ocean. *Bach, L.T., Ferderer, A.J., LaRoche, J., Schulz, K.G., 2024. Ocean Alkalinity Enhancement Pelagic Impact Intercomparison Project (OAEPIIP). Biogeosciences* 21 (16), 3665–3676.

Basso, D., Azzellino, A., Macchi, P., Santinelli, C., Fernández, E., Serret, P., Bachi, G., Checucci, G., Diaz, A., Teira, E., Raos, G., Valsecchi, S., Varliero, S., Bazzicalupo, P., Gariboldi, K., Gonzalez, J., 2023. Ocean liming in eutrophic water: a mesocosm scale approach. *EGU General Assembly*, 10342. <https://doi.org/10.5194/egusphere-egu23-10342>. Vienna, Austria, 24–28 Apr 2023, EGU23, 2023.

Basso, D., Bazzicalupo, P., Varliero, S., González, J., Serret, P., Pitta, P., Alcaráz, P., Penín, A., Macchi, P., Raos, G., Barbaccia, E., Magiopoulos, I., Tsiola, A., Romano, F., Valsecchi, S., Azzellino, A., 2024. Ocean liming in eutrophic vs. ultraoligotrophic environments and the response of algal calcifiers. *EGU General Assembly*, 13685. <https://doi.org/10.5194/egusphere-egu24-13685>, 2024, Vienna, Austria, 14–19 Apr 2024, EGU24.

Bhaumik, A., Faucher, G., Henning, M., Meunier, C.L., Boersma, M., 2025. Prey dynamics as a buffer: enhancing copepod resilience to ocean alkalinity enhancement. *Environ. Res. Lett.*

Biswas, H., Alexander, C.R.O.S., Yadav, K., Ramana, V.V., Prasad, V.R., Acharyya, T., Babu, P.R., 2011. The response of a natural phytoplankton community from the Godavari River Estuary to increasing CO2 concentration during the pre-monsoon period. *J. Exp. Mar. Biol. Ecol.* 407 (2), 284–293.

Boyd, P.W., 2019. Physiology and iron modulate diverse responses of diatoms to a warming Southern Ocean. *Nat. Clim. Change* 9, 148–152.

Calbet, A., Landry, M.R., 2004. Phytoplankton growth, microzooplankton grazing, and carbon cycling in marine systems. *Limnol. Oceanogr.* 49, 51–57.

Calbet, A., Martínez, R.A., Isari, S., Zervoudaki, S., Nejstgaard, J.C., Pitta, P., Sazhin, A. F., Sousoni, D., et al., 2012. Effects of light availability on mixotrophy and microzooplankton grazing in an oligotrophic plankton food web: evidences from a mesocosm study in Eastern Mediterranean waters. *J. Exp. Mar. Biol. Ecol.* 424–425, 66–77.

Calbet, A., Saiz, E., 2005. The ciliate-copepod link in marine ecosystems. *Aquat. Microb. Ecol.* 38, 157–167.

Camatti, E., Valsecchi, S., Caserini, S., Barbaccia, E., Santinelli, C., Basso, D., Azzellino, A., 2024. Short-term impact assessment of ocean liming: a copepod exposure test. *Mar. Pollut. Bull.* 198.

Castellani, C., Irigoien, X., Mayor, D.J., Harris, R.P., Wilson, D., 2008. Feeding of *Calanus finmarchicus* and *Oithona similis* on the microplankton assemblage in the Irminger Sea, North Atlantic. *J. Plankton Res.* 30, 1095–1116.

Chambers, J.M., 1992. Linear models. In: Chambers, J.M., Hastie, T.J. (Eds.), Chapter 4 of *Statistical Models in S*. Wadsworth & Brooks/Cole.

Claustre, H., Legendre, L., Boyd, P.W., Levy, M., 2021. The oceans’ biological carbon pumps: framework for a research observational community approach. *Front. Mar. Sci.* 8, 1–21.

Costanza, R., de Groot, R., Farber, S., Grasso, M., Hannon, B., Limburg, K., Van Den Belt, M., 1997. The value of the world’s ecosystem services and natural capital. *Ecol. Econ.* 25 (1), 3–15.

Cripps, G., Flynn, K.J., Lindeque, P.K., 2016. Ocean acidification affects the phyto-zooplankton trophic transfer efficiency. *PLoS One* 11.

Dolan, J.R., Claustre, H., Carlotti, F., Plounevez, S., Moutin, T., 2002. Microzooplankton Diversity: Relationships of Tintinnid Ciliates with Resources, Competitors and Predators from the Atlantic Coast of Morocco to the Eastern Mediterranean.

Doney, S.C., Fabry, V.J., Feely, R.A., Kleypas, J.A., 2009. Ocean acidification: the other CO2 problem. *Ann. Rev. Mar. Sci.* 1, 169–192.

Doney, S.C., Mitchell, K.A., Henson, S.A., Cavan, E., DeVries, T., Gruber, N., Hauck, J., Mouw, C.B., et al., 2024. Observational and numerical modeling constraints on the global ocean biological carbon pump. *Glob. Biogeochem. Cycles* 38.

Dorham, M., Abdel-Aziz, N., 2001. Abundance, species composition, biodiversity, size and ecology of tintinnins in Doha Harbour (Arabian gulf). *Environ. Sci. Biol. Arab Gulf J. Sci. Res.*

Ducklow, H., Steinberg, D., Buesseler, K., 2001. Upper ocean carbon export and the biological pump. *Oceanography* 14, 50–58.

Durbin, E.G., Campbell, R.G., Gilman, S.L., Dtrsb, A.G., 1995. Diel Feeding Behavior and Ingestion Rate in the Copepod *Calanus finmarchicus* in the Southern Gulf of Maine During Late Spring.

Durbin, A.G., Durbin, E.G., Włodarczyk, E., 1990. Diel Feeding Behavior in the Marine Copepod *Acartia tonsa* in Relation to Food Availability.

Feng, E.Y., Koeve, W., Keller, D.P., Oschlies, A., 2017. Model-based assessment of the CO2 sequestration potential of coastal ocean alkalization. *Earths Future* 5 (12), 1252–1266.

Ferderer, A., Chase, Z., Kennedy, F., Schulz, K.G., Bach, L.T., 2022. Assessing the influence of ocean alkalinity enhancement on a coastal phytoplankton community. *Biogeosciences* 19, 5375–5399.

Foteinis, S., Andresen, J., Campo, F., Caserini, S., Renforth, P., 2022. Life cycle assessment of ocean liming for carbon dioxide removal from the atmosphere. *J. Clean. Prod.* 370, 133309.

Friedlingstein, P., O’sullivan, M., Jones, M.W., Andrew, R.M., Gregor, L., Hauck, J., Le Quéré, C., Luijckx, I.T., et al., 2022. Global carbon budget 2022. *Earth Syst. Sci. Data* 14, 4811–4900.

Gagern, A., Manley, J., Kapsenberg, L., 2022. Ocean-based carbon dioxide removal: A new frontier in the blue economy. *Mar. Technol. Soc. J.* 56 (1), 40–48.

Guinotte, J.M., Fabry, V.J., 2008. Ocean acidification and its potential effects on marine ecosystems. *Ann. N. Y. Acad. Sci.* 1134, 320–342.

Guo, J.A., Strzepek, R.F., Swadling, K.M., Townsend, A.T., Bach, L.T., 2024. Influence of ocean alkalinity enhancement with olivine or steel slag on a coastal plankton community in Tasmania. *Biogeosciences* 21, 2335–2354.

Hammill, E., Johnson, E., Atwood, T.B., Harianto, J., Hinchliffe, C., Calosi, P., Byrne, M., 2018. Ocean acidification alters zooplankton communities and increases top-down pressure of a cubozoan predator. *Glob. Change Biol.* 24, e128–e138.

Hannides, C.C.S., Siokou, I., Zervoudaki, S., Frangoulis, C., Lange, M.A., 2015. Mesozooplankton biomass and abundance in Cyprus coastal waters and comparison with the Aegean Sea (Eastern Mediterranean). *Mediterr. Mar. Sci.* 16, 373–384.

Hartmann, J., Saitner, N., Lim, C., Schneider, J., Marín-Samper, L., Aristegui, J., et al., 2022. Stability of alkalinity in ocean alkalinity enhancement (OAE) approaches—consequences for durability of CO2 storage. *Biogeosci. Discuss.* 2022, 1–29.

Heneash, A.M.M., Abdel-Rahman, N.S., Gharib, S.M., 2015. Community composition, abundance and biomass of tintinnids (Ciliata: Protozoa) in the Western Harbour, south-eastern Mediterranean Sea, Egypt. *Environ. Monit. Assess.* 187.

Hofmann, G.E., Barry, J.P., Edmunds, P.J., Gates, R.D., Hutchins, D.A., Klinger, T., Sewell, M.A., 2010. The effect of ocean acidification on calcifying organisms in marine ecosystems: an organism-to-ecosystem perspective. *Annu. Rev. Ecol. Evol. Syst.* 41 (1), 127–147.

- Iglesias-Rodríguez, M.D., Rickaby, R.E.M., Singh, A., Gately, J.A., 2023. Laboratory Experiments in Ocean Alkalinity Enhancement Research.
- Iversen, M.H., 2023. Carbon Export in the Ocean: a Biologist's Perspective, 48, p. 56.
- Johnson, M., van Doorn, E., Hilmi, N., Marandino, C., McDonald, N., Thomas, H., Allemand, D., Algarin, L.D., et al., 2024. Can coastal and marine carbon dioxide removal help to close the emissions gap? Scientific, legal, economic, and governance considerations. *Elem. Sci. Anth.* 12.
- Lee, D.B., Song, H.Y., Park, C., Choi, K.H., 2012. Copepod feeding in a coastal area of active tidal mixing: Diel and monthly variations of grazing impacts on phytoplankton biomass. *Mar. Ecol.* 33, 88–105.
- Legendre, P., Legendre, Louis, 2012. Numerical Ecology. Elsevier.
- Longhurst, A.R., 1985. Relationship between diversity and the vertical structure of the upper ocean. *Deep-Sea Res., Part A* 32, 1535–1570.
- Mauchline, J., 1998. The biology of calanoid copepods. *Adv. Mar. Biol.* 33.
- Mayor, D.J., Sommer, U., Cook, K.B., Viant, M.R., 2015. The metabolic response of marine copepods to environmental warming and ocean acidification in the absence of food. *Sci. Rep.* 5.
- Mengis, N., Paul, A., Fernández-Méndez, M., 2023. Counting (on) blue carbon—Challenges and ways forward for carbon accounting of ecosystem-based carbon removal in marine environments. *PLOS Climate* 2, e0000148.
- Meyers, M.T., Cochlan, W.P., Carpenter, E.J., Kimmerer, W.J., 2019. Effect of ocean acidification on the nutritional quality of marine phytoplankton for copepod reproduction. *PLoS One* 14.
- Oschlies, A., Bach, L., Rickaby, R., Satterfield, T., Webb, R.M., Gattuso, J.P., 2023. Climate targets, carbon dioxide removal and the potential role of ocean alkalinity enhancement. *State Planet Discuss.* 2023, 1–11.
- Paul, J.A., Haunost, M., Goldenberg, S.U., Hartmann, J., Sánchez, N., Schneider, J., Suiter, N., Riebesell, U., 2024. Ocean alkalinity enhancement in an open ocean ecosystem: biogeochemical responses and carbon storage durability. *EGUsphere*.
- Pedersen, F., Hansen, P., 2003. Effects of high pH on a natural marine planktonic community. *Mar. Ecol. Prog. Ser.* 260, 19–31.
- Pedersen, S.A., Hansen, B.H., Altin, D., Olsen, A.J., 2013. Medium-term exposure of the North Atlantic copepod *Calanus finmarchicus* (Gunnerus, 1770) to CO₂-acidified seawater: effects on survival and development. *Biogeosciences* 10, 7481–7491.
- Pierce, R.W., Turner, J.T., 1992. Ecology of Planktonic Ciliates in Marine Food Webs.
- Pitta, P., Magiopoulos, I., Romano, F., Tsiola, A., Chantzaras, C., Gonzalez, J., Serret, P., Valsecchi, et al., 2024. Ocean liming in the oligotrophic Eastern Mediterranean: impact on the planktonic microbial food web. *EGU General Assembly*, 18538. <https://doi.org/10.5194/egusphere-egu24-18538>, 2024, Vienna, Austria, 14–19 Apr 2024, EGU24, 2024.
- Pitta, P., Nejstgaard, J.C., Tsagaraki, T.M., Zervoudaki, S., Egge, J.K., Frangoulis, C., et al., 2016. Confirming the “Rapid phosphorus transfer from microorganisms to mesozooplankton in the Eastern Mediterranean Sea” scenario through a mesocosm experiment. *J. Plankton Res.* 38 (3), 502–521.
- Pitta, P., Giannakourou, A., Christaki, U., 2001. Planktonic ciliates in the oligotrophic Mediterranean Sea: longitudinal trends of standing stocks, distributions and analysis of food vacuole contents. *Aquat. Microb. Ecol.* 24, 297–311.
- R Core Team, 2019. R: a Language and Environment for Statistical Computing. R Foundation for statistical computing, Vienna, Austria. <https://www.R-project.org/>.
- Richardson, T.L., Jackson, G., 2007. Small phytoplankton and carbon export from the surface ocean. *Science* 315, 838–840.
- Riebesell, U., Aberle-Malzahn, N., Achterberg, E.P., Algueró-Muñiz, M., Alvarez-Fernandez, S., Aristegui, J., Bach, L.T., Boersma, M., et al., 2018. Toxic algal bloom induced by ocean acidification disrupts the pelagic food web. *Nat. Clim. Change* 8, 1082–1086.
- Serret, P., Basso, D., Pitta, P., Magiopoulos, I., Alcaraz, P., Penín, A., Tsiola, A., Romano, et al., 2024. The impact of ocean liming on phytoplankton size-structure and the balance of photosynthesis and respiration in two contrasting environments. *EGU General Assembly*, 13093. <https://doi.org/10.5194/egusphere-egu24-13093>, 2024, Vienna, Austria, 14–19 Apr 2024, EGU24, 2024.
- Shannon, C.E., Weaver, W., 1949. The Mathematical Theory of Information. University of Illinois Press, Urbana, p. 97.
- Siegel, D.A., Burd, A.B., Estapa, M., Fields, E., Johnson, L., Romanelli, E., Brzezinski, M. A., Buesseler, K.O., et al., 2024. Dynamics of aggregates and sinking carbon fluxes in a. *Proc. Natl. Acad. Sci.*
- Siegel, D.A., Devries, T., Doney, S.C., Bell, T., 2021. Assessing the sequestration time scales of some ocean-based carbon dioxide reduction strategies. *Environ. Res. Lett.* 16.
- Siokou, I., Frangoulis, C., Grigoratou, M., Pantazi, M., 2014. Zooplankton community dynamics in the N. Aegean front (E. Mediterranean) in the winter-spring period. *Mar. Sci.* 15, 706–720.
- Steinberg, D.K., Landry, M.R., 2017. Zooplankton and the ocean carbon cycle. *Ann. Rev. Mar. Sci.* 9, 413–444.
- Subhas, A.V., Marx, L., Reynolds, S., Flohr, A., Mawji, E.W., Brown, P.J., Cael, B.B., 2022. Microbial ecosystem responses to alkalinity enhancement in the North Atlantic Subtropical Gyre. *Front. Clim.* 4.
- Suiter, N., Faucher, G., Lim, C., Schneider, J., Moras, C.A., Riebesell, U., Hartmann, J., 2024. Ocean alkalinity enhancement approaches and the predictability of runaway precipitation processes: results of an experimental study to determine critical alkalinity ranges for safe and sustainable application scenarios. *Biogeosciences* 21 (20), 4587–4604.
- Swadling, K.M., Slotwinski, A., Davies, C., Beard, J., McKinnon, A.D., Coman, F., Murphy, N., Tonks, M., Rochester, W., Conway, D.V.P., Hosie, G.W., Richardson, A. J., 2013. Australian Marine Zooplankton: a Taxonomic Guide and Atlas. February 2013, Version 1.0.
- Tamme, E., Beck, L.L., 2021. European carbon dioxide removal policy: current status and future opportunities. *Front. Clim.* 3.
- Tilbrook, B., Jewett, E.B., DeGrandpre, M.D., Hernandez-Ayon, J.M., Feely, R.A., Gledhill, D.K., Hansson, L., Isensee, K., et al., 2019. An enhanced ocean acidification observing network: from people to technology to data synthesis and information exchange. *Front. Mar. Sci.* 6.
- Turner, J.T., 2015. Zooplankton fecal pellets, marine snow, phytodetritus and the ocean's biological pump. *Prog. Oceanogr.* 130, 205–248.
- Weydmann, A., Soreide, J.E., Kwasniewski, S., Widdicombe, S., 2012. Influence of CO₂-induced acidification on the reproduction of a key Arctic copepod *Calanus glacialis*. *J. Exp. Mar. Biol. Ecol.* 428, 39–42.
- Wiadnyana, N., Rassoulzadegan, F., Wiadnyana, N.N., Rassoulzadegan, Fereidoun, 1989. Selective feeding of *Acartia clausi* and *Centropages typicus* on microzooplankton. *Mar. Ecol. Prog. Ser.* 53, 37–45.
- Zervoudaki, S., Christou, E.D., Nielsen, T.G., Siokou-Frangou, I., Assimakopoulou, G., Giannakourou, A., Maar, M., Pagou, K., et al., 2007. The importance of small-sized copepods in a frontal area of the Aegean Sea. *J. Plankton Res.* 29, 317–338.
- Zervoudaki, S., Sakavara, A., Protopapa, M., Christou, E., Siokou, I., 2020. Mesozooplankton Dynamics in the Aegean Sea, pp. 115–152.